

$$f(x) = 5x^3 - 3x^2 + 1$$

$$a_0 = \frac{1}{\pi} \int_{-\pi}^{\pi} 5x^3 - 3x^2 + 1 \, dx$$

$$= \frac{1}{\pi} \left[\frac{5x^4}{4} - x^3 + x \right]_{-\pi}^{\pi}$$

$$= \frac{1}{\pi} \left[\frac{5(\pi)^4}{4} - (\pi)^3 + (\pi) - \left[\frac{5(-\pi)^4}{4} - (-\pi)^3 + (-\pi) \right] \right]$$

$$= \frac{1}{\pi} \left[\frac{5\pi^4}{4} - \pi^3 + \pi \right] - \left[\frac{5\pi^4}{4} + \pi^3 - \pi \right] = -2\pi^2 + 2$$

$$\frac{a_0}{2} = \frac{-2\pi^2 + 2}{2} = -\pi^2 + 1 = \underline{\underline{1 - \pi^2}}$$

$$b_n = \frac{1}{\pi} \int_{-\pi}^{\pi} (5x^3 - 3x^2 + 1) \sin(nx) \, dx$$

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$$(-3x^2 + 1) \sin(nx) \rightarrow \text{impar} = 0$$

$\therefore$

$$b_n = \frac{1}{\pi} \int_{-\pi}^{\pi} 5x^3 \sin(nx) \, dx$$

$$\begin{array}{lcl} 5x^3 & \rightarrow & \sin(nx) \\ 15x^2 & \rightarrow & -\frac{\cos(nx)}{n} \\ 30x & \rightarrow & -\frac{\sin(nx)}{n^2} \\ 30 & \rightarrow & \frac{\cos(nx)}{n^3} \\ 0 & \rightarrow & \frac{\sin(nx)}{n^4} \end{array}$$

$$= \frac{1}{\pi} \left[-\frac{5x^3}{n} \cos(nx) + \frac{15x^2}{n^2} \sin(nx) + \frac{30x}{n^3} \cos(nx) - \frac{30}{n^4} \sin(nx) \right]_{-\pi}^{\pi}$$

$$= \frac{1}{\pi} \left[-\frac{5x^3}{n} \cos(nx) + \frac{30x}{n^3} \cos(nx) \right]_{-\pi}^{\pi}$$

$$= \frac{1}{\pi} \left[\left(-\frac{5\pi^3}{n} + \frac{30\pi}{n^3} \right) (-1)^n \right] - \left[\left(\frac{5\pi^3}{n} - \frac{30\pi}{n^3} \right) (-1)^n \right] =$$

$$= \frac{1}{\pi} \left[2 \left(-\frac{5\pi^3}{n} + \frac{30\pi}{n^3} \right) (-1)^n \right] = \left[\left(-\frac{10\pi^3}{n} + \frac{60\pi}{n^3} \right) (-1)^n \right] \frac{1}{\pi}$$

$$= \frac{1}{\pi} \left(-\frac{10\pi^3}{n} + \frac{60\pi}{n^3} \right) (-1)^n = \left(-\frac{10\pi^2}{n} + \frac{60}{n^3} \right) (-1)^n = \frac{10(-1)^{n+1} (\pi^2 n^2 - 6)}{n^3} //$$

$$a_n = \frac{1}{\pi} \int_{-\pi}^{\pi} (-3x^2 + 1) \cos(nx) dx$$

$$\begin{array}{lcl} -3x^2 + 1 & \searrow & \cos(nx) \\ -6x & \searrow & \frac{\sin(nx)}{n} \\ -6 & \searrow & \frac{\cos(nx)}{n^2} \\ 0 & \searrow & -\frac{\sin(nx)}{n^3} \end{array}$$

$$= \frac{1}{\pi} \left[\left(-3x^2 + 1 \right) \frac{\sin(nx)}{n} - 6x \frac{\cos(nx)}{n^2} + 6 \frac{\sin(nx)}{n^3} \right]_{-\pi}^{\pi}$$

$$= \frac{1}{\pi} \left[\left. -\frac{6x \cos(nx)}{n^2} \right|_{-\pi}^{\pi} \right] = \left[\left. -\frac{6\pi \cos(n\pi)}{n^2} \right] - \left[\left. -\frac{6(-\pi) \cos(-n\pi)}{n^2} \right] \right] \frac{1}{\pi}$$

$$= \frac{1}{\pi} \left[\left(-\frac{6\pi}{n^2} (-1)^n \right) - \left(\frac{6\pi}{n^2} (-1)^n \right) \right] = \left[-\frac{12\pi}{n^2} (-1)^n \right] \frac{1}{\pi} = -\frac{12(-1)^n}{n^2}$$